EXPERIMENT :- 1

PROGRAM CODES ➖

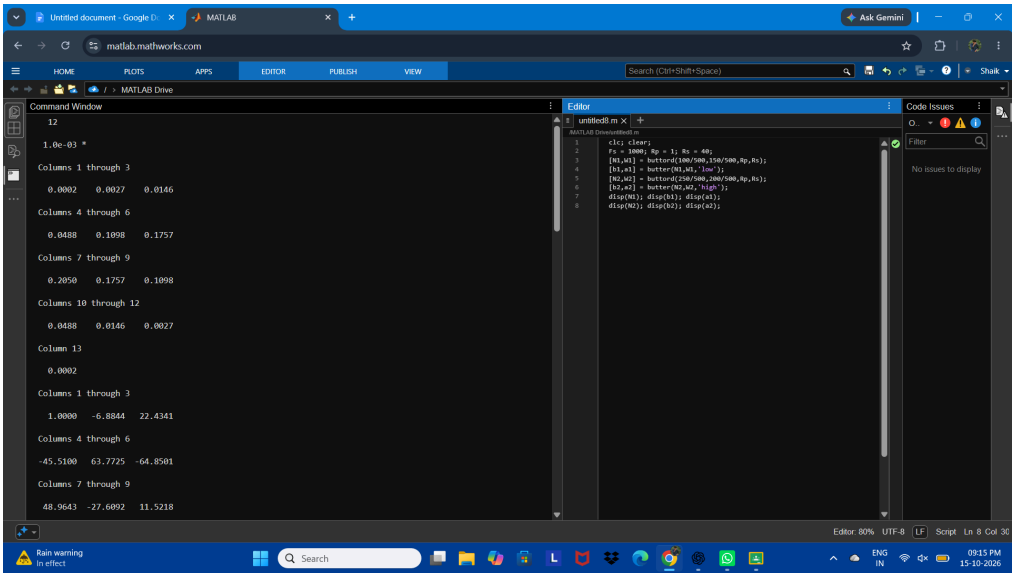
```
>> clc; clear; close all;
Fs = 1000;
[b1,a1] = butter(4,100/(Fs/2),'low');
[b2,a2] = butter(4,250/(Fs/2),'high');
freqz(b1,a1,512,Fs);
figure; freqz(b2,a2,512,Fs);
>>
```

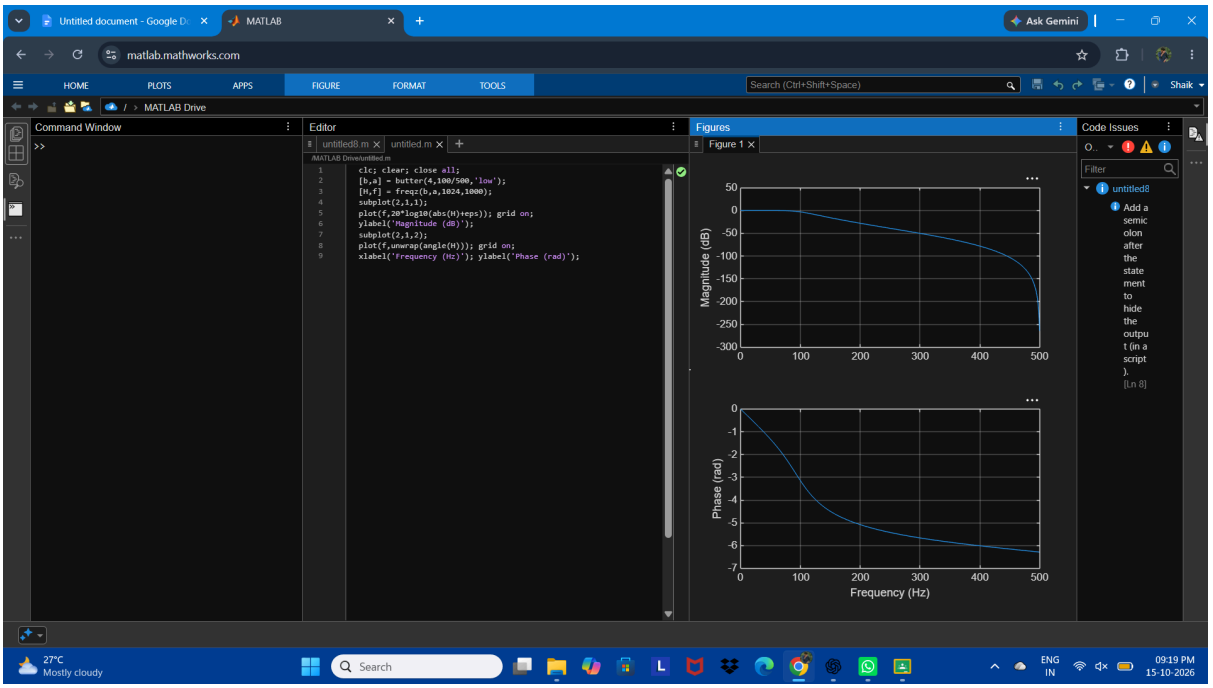
```
clc; clear;
Fs = 1000; Rp = 1; Rs = 40;
[N1,W1] = buttord(100/500,150/500,Rp,Rs);
[b1,a1] = butter(N1,W1,'low');
[N2,W2] = buttord(250/500,200/500,Rp,Rs);
[b2,a2] = butter(N2,W2,'high');
disp(N1); disp(b1); disp(a1);
disp(N2); disp(b2); disp(a2);
```

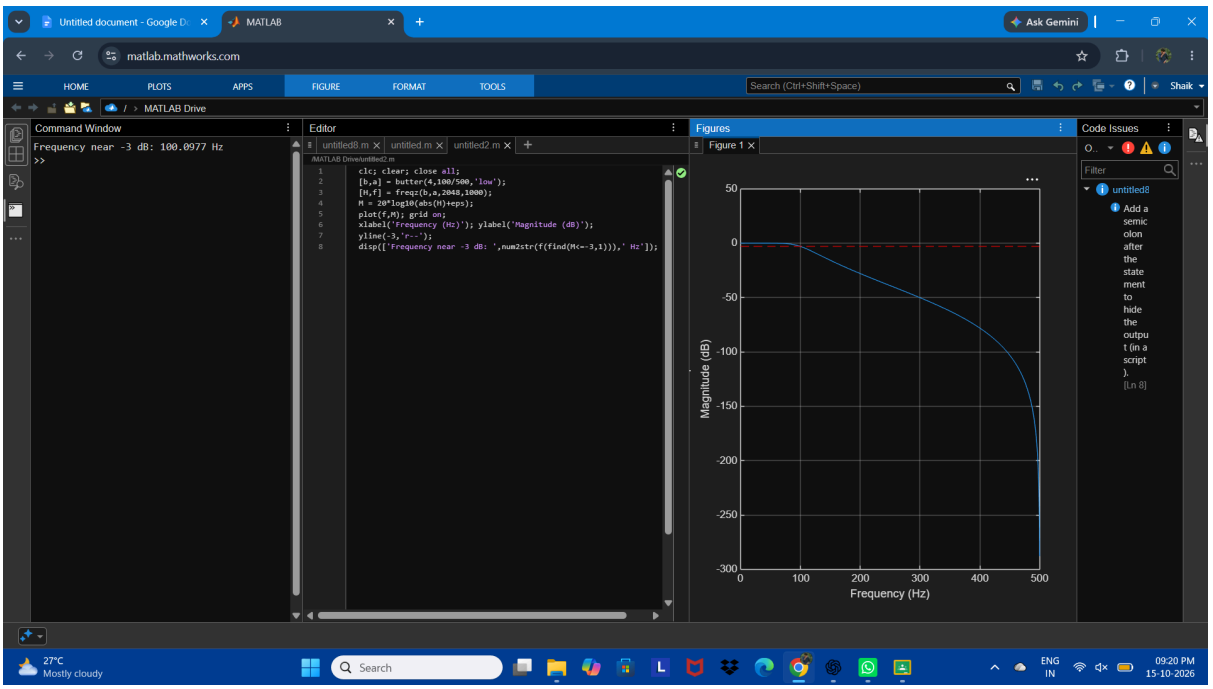
```
clc; clear; close all;
[b,a] = butter(4,100/500,'low');
[H,f] = freqz(b,a,1024,1000);
subplot(2,1,1);
plot(f,20*log10(abs(H)+eps)); grid on;
ylabel('Magnitude (dB)');
subplot(2,1,2);
plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
```

```
clc; clear; close all;
[b,a] = butter(4,100/500,'low');
[H,f] = freqz(b,a,2048,1000);
M = 20*log10(abs(H)+eps);
plot(f,M); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
yline(-3,'r--');
disp(['Frequency near -3 dB: ',num2str(f(find(M<=-3,1))),' Hz']);
```






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clc; clear; close all;
[b,a] = butter(4,100/500,'low');
[H,f] = freqz(b,a,1024,1000);
subplot(2,1,1);
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ylabel('Magnitude (dB)');
subplot(2,1,2);
plot(f,unwrap(angle(H))); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
Figures
Figure 1
Magnitude (dB)
Phase (rad)
Frequency (Hz)
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clc; clear; close all;
[b,a] = butter(4,100/500,'low');
[H,f] = freqz(b,a,2048,1000);
M = 20*log10(abs(H)+eps);
plot(f,M); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
yline(-3,'r--');
disp(['Frequency near -3 dB: ',num2str(f(find(M<=-3,1))),' Hz']);
Figures
Figure 1
Magnitude (dB)
Frequency (Hz)
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